

MS5612: Real Options Valuation

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Contents

1	27th July - Monday	3
1.1	DCF vs Options Approach	3
2	29th July - Wednesday	3
2.1	NPV	4
2.2	Class Discussion on Strategic Investment	4
2.3	Example	4
2.3.1	Case 1: Cost Certain, Revenue uncertain	4
2.3.2	Case 2: Revenue certain at 90, cost uncertain	4
2.3.3	Case 3: Revenue and cost are uncertain	5

1 27th July - Monday

Finance could broadly be divided into 3 parts: 1. Personal Finance, Corporate Finance, and Financial Markets

How do businesses manage their financing? - Corp Fin is mostly about it.

Corp Fin could again broadly be classified into two: Investment and Financing.

Investment decisions result in assets and financing decisions result in liabilities.

Stocks could be valued in any of the following methods:

- Comparative method
- Dividend Discount method
- Free Cash Flow method
- Book Value

How do we value Debt?

- present value of cashflows of the debt (interest + principal)

How do we value projects (assets)?

- Discounted Cashflow Method (DCM)

Keyword: Fortune 500 Keyword: Managers **create** shareholder value.

Most Fortune 500 firms use real options. But as per McKinzie's India head, almost no Indian company is using real options. So, there is an opportunity.

1.1 DCF vs Options Approach

- DCF involves limited discretionary power, but options provide flexibility.
- If people are liking flexibility, we may value this flexibility.
- Managers create flexibility where seemingly none exists.

2 29th July - Wednesday

Focus of the course is on the management of long term assets (not much on liabilities).

NPV, IRR, Profitability Index (PI), Payback Period.

NPV - How much does the project worth? (**How much?**)

$$NPV = \sum_{i=1}^n \frac{C_i}{(1+r)^i} - IO$$

Keyword: IO - Initial Outlay

IRR - What returns are we making? (**What percent?**)

PI - Present value of cash inflow/ initial outlay (IO) (**What rank?**)

PB - **How long?**

2.1 NPV

$$NPV = \sum_{i=1}^n \frac{C_i}{(1+r)^i} - IO$$

Investment rule: if $NPV > 0$ we may accept
if $NPV < 0$ we must reject
if $NPV = 0$ we reject it as it doesn't create any shareholder value.

2.2 Class Discussion on Strategic Investment

What are Strategic Investments?

How do you justify an Investment? NPV, IRR, PB, PI? But only these, really?

Some investments could have negative NPV, but they could still be chosen.

Strategic investments are those that may fail the test of conventional investment tests (NPV, IRR) but they may pass the tests in different cases.

2.3 Example

Let's say there's an RnD project requiring 15 million USD.

To manufacture the plant, the cost could be 40 million, 80 million, and 120 million.

And the market share could be 50 million, or 130 million

The expected cost is 80 million, expected benefit is 90 million, and 15 millions for RnD.

Here, conventionally, the recommendation is to not invest and the NPV is -5 million USD.

2.3.1 Case 1: Cost Certain, Revenue uncertain

Let's say once we've invested in RnD, the manufacturing cost is certainly 80 million. If, now the market is 50 million, we do not invest. If however the market is 130 million, we go ahead and invest, in which case the benefit is 50 million. NPV in this case is $\frac{1}{2}(0 + 15) - 15 = 10$ million.

2.3.2 Case 2: Revenue certain at 90, cost uncertain

if cost is 40 mil, we invest, benefit = 50 mil
if cost is 80 mil, we invest, benefit = 10 mil
if cost is 120 mil, we do not invest, benefit = 0

$$NPV = \frac{1}{3}(50 + 10 + 0) - 15 = 5 \text{ million.}$$

2.3.3 Case 3: Revenue and cost are uncertain

$(40, 50) = 10$ $(40, 130) = 90$ $(80, 50) = 0$ $(80, 130) = 50$ $(120, 50) = 0$ $(120, 130) = 10$

$NPV = 1/6(10 + 90 + 50 + 10) - 15 = 11.67$ million

In the formula of NPV, r is the measure of risk.

The discount rate for a project must reflect the risk of the project.

Case 3 is definitely the riskiest, but what about the other cases? Which one is more risky? We can calculate the risk? We may also calculate the variance. We'll notice that the NPV increases as the risk increases.

So, this RnD investment is an option that would give the firm the flexibility to go ahead and set up a manufacturing plant. And the riskier the option is, the more valuable it is.